AMRITA VISHWA VIDYAPEETHAM
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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I124*

Team Name: *IntelliHive*

Domain: *Climate Action, Water & Environmental Resilience*

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Presentation Outline

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Problem Statement

Rainfall forecasts tell us **how much rain** is coming, but not **which street will drown, how deep, or when**. Without a street-level warning, cities react only after the water rises, and the cost is paid in **gridlock, damaged property and lives**.

Which street?

How deep?

When?

Today's forecasts cannot answer these three questions.

Problem Understanding

Who is affected

- Residents and commuters in cities like Chennai
- City corporations
- Emergency services and traffic police

Cause

- Heavy rain
- Concrete surfaces and low-lying ground
- Drains that are full or blocked

Effect

- Traffic jams
- Damaged property
- Delayed ambulances and risk to life

Existing Gaps & Challenges

Existing approach	What is missing
Forecasts and radar	Show rainfall, not street-level flood depth
Drain and surface models (Yang et al., 2020)	Accurate, but need heavy data and are slow for live use
ML nowcasting (Farahmand et al., 2023)	Fast, but area-level and without the drain network

Unsolved: in our literature review we found no study that links nowcasting, a drain network and explainable ML in one workflow for Indian cities.

Our Proposed Solution

Urban Flood Guardian: predict, explain, act

1. Predict

- Flood depth (cm) for each street, 0–3 hours ahead
- Risk level and time to critical

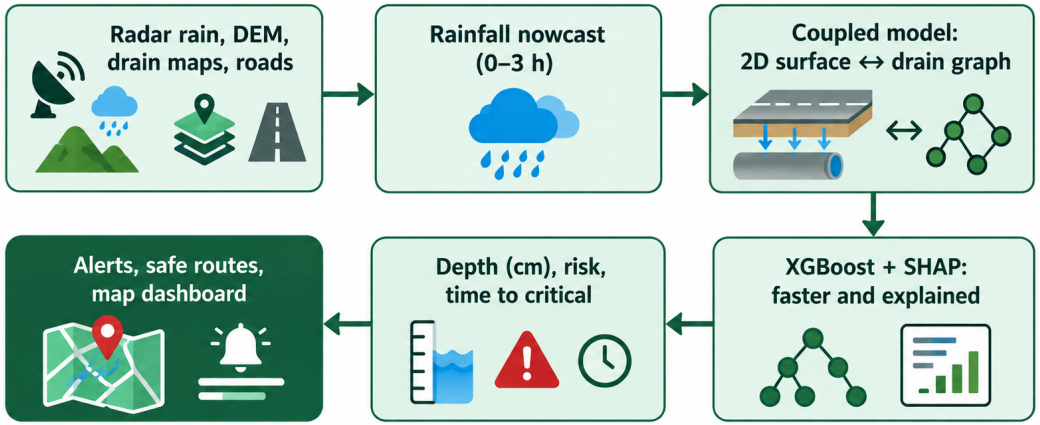
2. Explain

- Rain, terrain and drain overflow in one model
- XGBoost + SHAP shows why a street is at risk

3. Act

- Alerts when depth crosses a safe limit
- Flood-safe routes
- Map for authorities

Technical Approach



IMD gives rainfall for an area. Terrain, surface type and drain capacity turn it into flood depth for each street.

We calibrate on past floods and show a range, not one value.

Rainfall is the input; the coupled model and the ML layer turn it into depth, risk and action for each street.

Tech Stack

Frontend: React, MapLibre GL

Backend: FastAPI

Database: PostgreSQL + PostGIS

AI/ML, APIs, or Hardware: pySTEPS, NetworkX, XGBoost, SHAP, FCM/SMS

Innovation & Uniqueness

What is new

- Rain, terrain and drain overflow in one model
- Every prediction comes with a reason

Why it fits

- Floods are very local
- Drains matter as much as rain
- Rain-only forecasts miss the cause

Differentiator

- Predict, explain, prioritise and act
- Time to critical and safe routes

Feasibility & Scalability

Feasibility

- Start with a few Chennai hotspots
- Existing data: IMD radar, DEM, GCC drain maps, OpenStreetMap
- Open-source tools

Scalability

- Expand zone by zone
- Same pipeline for other cities
- Works in any city with rainfall, terrain and drain data

Risks and fixes

- Missing drain data: use proxy values, replace later
- Accuracy drops towards the 3 h end: show a range
- Heavy computing: ML surrogate
- Rainfall is per area, not per street: blend radar with gauges, calibrate on past floods

Cost for Product (Optional)

Development

- Open-source software (Python, PostGIS, React)
- Mainly team time

Hosting

- Small cloud server
- To be estimated for the pilot area

Estimated total

- To be confirmed once the pilot scope is fixed

Use Cases

1. Ambulance

- **Dispatcher** gets a flood-safe route that avoids streets expected to go deep in the next hour

2. Engineer

- **Municipal engineer** sees which drains are about to overflow and sends crews before the rain peaks

3. Resident

- **Resident** gets an alert and an alternative route, and moves the vehicle out of a low street in time

Impacts and Benefits

Users

- Citizens: early warnings and safer routes
- Authorities: ranked list of roads and drains

Wider impact

- Fewer stuck vehicles, less damage
- Faster emergency response
- Proactive climate adaptation

Measurable outcome

- Warning lead time
- Depth error (cm)
- Flooded streets correctly flagged, tested on past floods

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 11	Target 11.5	Sustainable Cities and Communities	Warns before floods to cut losses and risk to life
SDG 13	Target 13.1	Climate Action	Builds resilience to flooding from heavier rain

Individual Member Contribution

Member	Work done so far
Archand S	Defined the problem; designed the coupled flood model and the ML layer
Kruthic Kumar N L	Identified data sources; studied drain network modelling and data gaps
Pranav S	Designed the architecture and tech stack; planned alerts and routing
Swetha T	Ran the literature search; found the research gap; prepared references

Yukti ID

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